ENGR489 PROJECT · VICTORIA UNIVERSITY OF WELLINGTON

HUNTING MR. X
WELLINGTON EDITION

A new Open Day demo:

a classic board game, played online on a real map of Wellington

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KEY FINDINGS

Future students

a demo for prospective students
at the university's Open Days

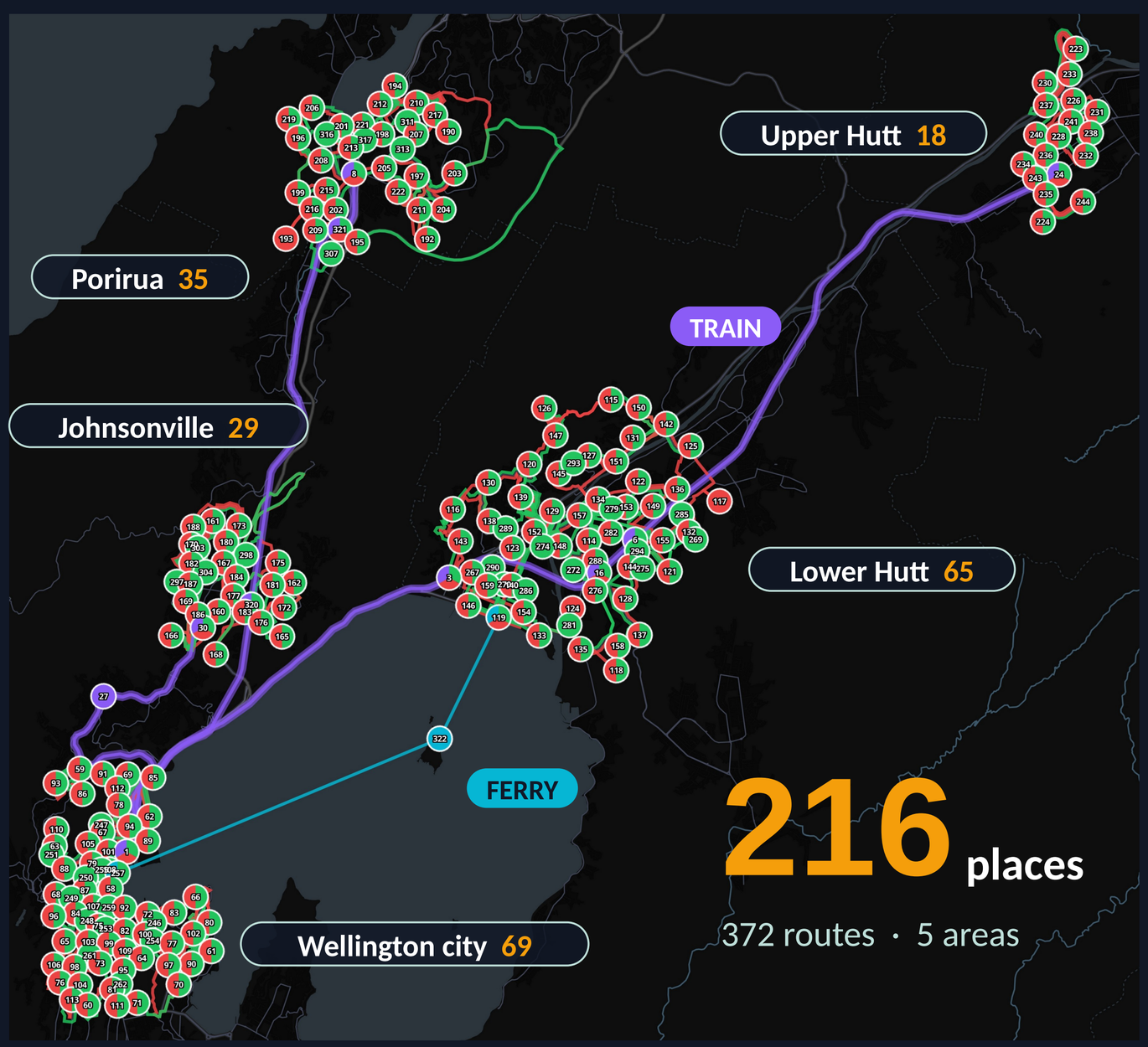
Scotland Yard

inspired by the classic board game
by Ravensburger

No install

join from a web browser
with a link or a six-character code

THE BOARD: ALL OF WELLINGTON



WHY OPEN DAY?

- The old software demo was over five years old
- Visitors need something to try in minutes
- A map of their own city is instantly familiar
- It shows what a fourth-year student can build

HOW TO PLAY

- Mr X hides; up to five detectives hunt him
- Every move spends a travel ticket
- Mr X is shown only every few rounds
- Catch him within 24 rounds to win

Escooter · Bus · Train · Ferry · Invisible · Double

PLAY IT NOW
Scan or visit
vuw-mrx.xyz

TECHNICAL CHALLENGES

KEEPING A SECRET

Each player is sent their own view, so Mr X's position never reaches a detective's screen.

STAYING IN SYNC

Every move is pushed live to every screen. A dropped connection catches up by itself.

A FREE REAL MAP

Routes along real roads are worked out once, not paid for on every move.

PROVING IT WORKS

The rules sit apart from the server, so each is tested exactly. 300 random games hunt for leaks.

TECH STACK

Screens	Vue 3 · TypeScript · Tailwind CSS
Map	MapLibre GL · OpenStreetMap
Server	Java 21 · Spring Boot
Live	WebSockets · STOMP
Testing	JUnit 5 · Mockito · PIT
Hosting	Docker · Cloudflare Tunnel

EVALUATION & OUTCOME

135	99.2%	100%	2
automated tests, all passing	of server code lines covered	of 199 planted bugs caught	rounds of user testing

OUTCOME

- Played live at Open Day
- Anyone can join over the internet
- Ready for future Open Days

NEXT STEPS

- A big-screen view for the stand
- A phone-friendly layout
- Computer players for solo games

Board-game mechanics used with permission from Ravensburger, for non-commercial academic use.